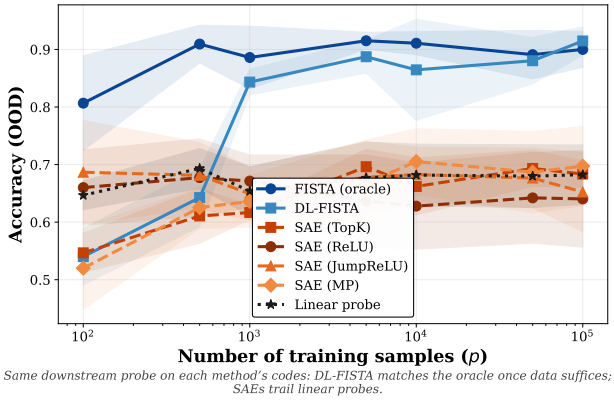

Figure 11: **More data closes the gap for DL-FISTA but not for SAEs.** OOD accuracy of a logistic probe on each method's codes. DL-FISTA matches FISTA (oracle) once $p \geq 10^3$ (0.88 vs 0.68 for the linear probe). All SAE variants trail the linear probe regardless of data. $d_z = 100$, $k = 10$, $d_y = 47$.

dictionary, initialised from $\mathbf{h}^{(0)} = \mathbf{0}$.

3. **Refined** (warm-start): FISTA on the same frozen $\hat{\mathbf{W}}$, initialised from $\mathbf{h}^{(0)} = r(\mathbf{y})$.

All three use the *same* dictionary—only the inference procedure differs. Figure 8 shows the comparison alongside DL-FISTA (which learns its own dictionary independently). The result shows that SAE encoders ≈ frozen FISTA ≈ refined (∼ 0.15–0.4 MCC), while DL-FISTA (which learns its own dictionary via classical alternating minimisation) achieves an MCC of ∼ 0.75–0.95. This implies that swapping the encoder for per-sample inference on the same dictionary does not meaningfully improve recovery. The SAE-learned dictionary itself is the binding constraint. This conclusion is robust to the choice of FISTA regularisation strength λ: sweeping λ across three orders of magnitude yields at most modest improvements over the SAE encoder, far below the oracle at every λ (appendix, Section D.7). Note that DL-FISTA also degrades at large d_z (to ∼0.3 at $d_z = 10$K, Section 4.2), reflecting the inherent difficulty of dictionary learning at scale—but it degrades uniformly across ID and OOD, unlike SAEs which exhibit a compositional generalisation failure. The linear probe, operating directly on $\mathbf{y}$ without sparse inference, degrades in MCC even faster as superposition intensifies (Figure 6), but maintains surprisingly high downstream accuracy (Figure 7) because it is supervised for that specific task—a divergence that illustrates how single-task metrics can mask identifiability failure.

At 100 FISTA iterations (sufficient for convergence of the convex Lasso objective) warm-starting from SAE codes yields the same solution as cold-starting from zeros, confirming that the per-sample optimisation converges regardless of initialisation. The one exception is TopK, whose SAE decoder achieves ∼0.87 OOD MCC on its own (comparable to DL-FISTA at small d_z), consistent with TopK's structurally enforced sparsity producing a more faithful dictionary.

The dictionary fails because the columns point in the wrong directions. We track cosine similarity between decoder columns and ground truth during training (Figure 12) and decompose column-level errors at convergence (Figure 28). At $d_z = 100$, DL-FISTA converges to high cosine (> 0.9) while SAEs plateau earlier—the SAE optimisation landscape is harder even when the problem is tractable for alternating minimisation. At $d_z = 5000$, neither method finds the right directions (cosine ∼0.2–0.35), implying that the SAE dictionaries do not overfit or drift, they simply never converge. Norm ratios remain ≈1.0 throughout, and re-normalising columns or substituting oracle norms does not improve MCC (Figure 27). TopK is an outlier at small d_z (cosine 0.93, 90% of columns close to ground truth at $d_z = 100$), but this advantage vanishes at scale.

Why the encoder fails: wrong support. We measure support recovery—whether the SAE activates the correct features—by comparing the binary nonzero pattern of SAE codes against the ground truth (after Hungarian matching). At $d_z = 5000$ with true sparsity $k = 10$: ReLU activates ∼120 of 5000 features (precision 0.009); JumpReLU activates ∼300 (precision 0.005). Even TopK, which activates the correct number of features (∼10), selects almost entirely wrong atoms (precision 0.03). Re-estimating magnitudes via least-squares on the SAE's support makes things *worse* (Figure 56), confirming the support itself is incorrect. At smaller d_z, TopK's support precision is substantially better (0.49 at $d_z = 100$; Figure 29), explaining its stronger performance at small scale.

Takeaway. The gap between oracle and other methods seems to be a dictionary learning problem. SAEs fail at both dictionary learning and inference, but the dictionary is the binding constraint: per-sample inference cannot rescue wrong column directions. At scale ($d_z = 5000$), DL-FISTA also fails at dictionary learning (cosine ∼0.25), converging to similar MCC as SAEs.

4.5 THE SAE DICTIONARY IS A USEFUL STARTING POINT

Although the SAE decoder is a poor standalone dictionary, it may still encode useful structure that accelerates dictionary learning. We test this by using the SAE decoder as the initial dictionary for DL-FISTA (unsupervised alternating optimisation), comparing against DL-FISTA initialised from a random dictionary.

Figure 13 shows convergence curves at $d_z = 100$, sweeping the number of dictionary-update rounds. The SAE decoder provides a clear head start: for TopK, warm-starting begins at OOD MCC 0.87 and reaches 0.94 within 5 rounds, while random initialisation starts at 0.30 and requires ∼50 rounds

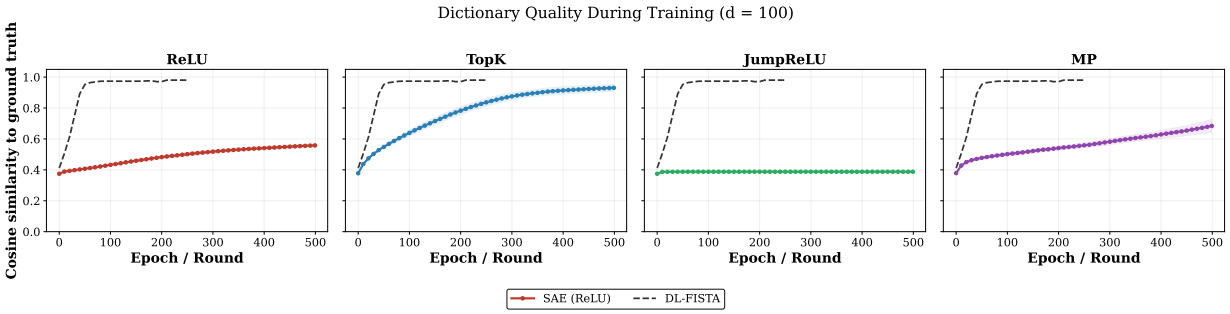

Figure 12: **DL-FISTA learns correct dictionary directions; SAEs do not.** SAE decoder cosine to ground truth vs training epoch (coloured) and DL-FISTA dictionary cosine vs outer round (gray dashed) at $d_z = 100$. At large d_z (≥ 5000), neither method converges (Figure 36).

to reach the same level. For ReLU and JumpReLU, the advantage is smaller but consistent—the SAE decoder saves ~20–50 rounds of dictionary learning. Both initialisations converge to the same final MCC, confirming that the SAE decoder biases the optimisation toward the correct basin without trapping it in a suboptimal one.

Encoder warm-start is marginal. We also test whether the SAE encoder's output provides a useful initialisation for per-sample FISTA (with frozen dictionary). Since the Lasso objective is convex, cold-start and warm-start must converge to the same optimum given sufficient iterations. At 100 iterations, both yield identical MCC. At low iteration budgets (1–10), warm-starting from SAE codes provides a modest advantage for TopK but negligible benefit for other types (appendix, Figure 26). The practical value of the SAE encoder as a FISTA initialiser is limited.

Takeaway. At small d_z, the SAE decoder is a useful warm-start for DL-FISTA, saving ~50 dictionary-update rounds. At large d_z (≥ 5000), this advantage vanishes—neither initialisation leads to a good dictionary. The practical path forward requires better dictionary learning algorithms that scale, not just better initialisations.

5 CONCLUSION

In this paper, we asked two questions: is sparse inference necessary under superposition, and if so, what is the bottleneck?

Sparse inference is necessary. Under superposition, concepts are linearly *represented* in activations but not linearly *accessible*—decision boundaries that are linear in latent space become nonlinear after projection. Linear probes, even with oracle access to ground-truth latents, degrade sharply in identifiability as superposition intensifies (MCC < 0.1 at $d_z = 10{,}000$). Per-sample FISTA with the ground-truth dictionary achieves near-perfect OOD recovery (MCC ≥ 0.83, accuracy ≥ 0.97) at all scales, proving the problem is solvable within the compressed-sensing framework.

The bottleneck is dictionary learning, not inference. SAE-learned decoder columns point in substantially wrong directions, and replacing the encoder with per-sample FISTA on the same dictionary does not help—the dictionary itself is the binding constraint. On the same downstream task, in our experiments, SAE codes offer no advantage over probing raw activations. Classical dictionary learning (DL-FISTA) dominates linear probes when it succeeds ($d_z \leq 1{,}000$). The gap between oracle and the best unsupervised method identifies **scalable dictionary learning** as the key open problem for sparse inference under superposition.

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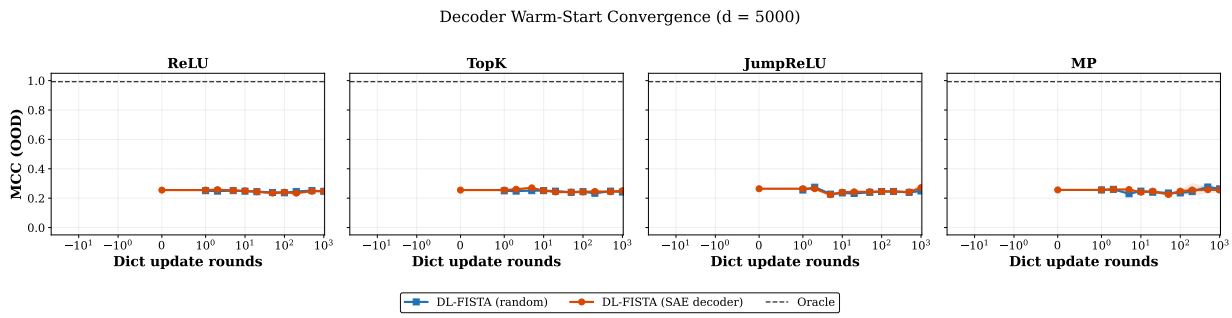

Figure 13: **The SAE decoder is a useful warm-start for dictionary learning.** Orange: DL-FISTA initialised from SAE decoder. Blue: DL-FISTA from random dictionary. Gray dashed: oracle. The SAE decoder provides a clear head start, especially for TopK, but both initialisations converge to the same optimum. $d_z = 5000$, $k = 10$. Results at other d_z in appendix.

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